

**AMENDMENT NO. 1 JULY 1992
TO
IS 9300 (Part 1) : 1979 STATISTICAL MODELS FOR
INDUSTRIAL APPLICATIONS**

PART 1 DISCRETE MODELS

(Page 5, clause 3.1.1, last line) — Substitute 'or their collection is known as event' for 'are known as events'

(Page 5, clause 3.1.2) — Substitute the following for the existing clause: —

'3.1.2 Exhaustive Event — The collection of all possible outcomes in any trial is known as exhaustive event'

(Page 5, clause 3.1.4) — Substitute the following for the existing clause.

'3.1.4 Favourable Event — The set of outcomes favourable to an event in a trial is the set of all outcomes which entail the happening of the event.'

(Page 5, clause 3.1.6) — Substitute the following for the existing clause:

'3.1.6 Independent Events — Several events are said to be independent if the chance of occurrence (or non-occurrence) of an event is not affected by the supplementary knowledge concerning the occurrence (or non-occurrence) of any of the remaining events'

(Page 6, clause 3.3.1) — Shift the matter of 3.4 at the end of this clause.

(Page 7, clause 3.3.2) — Shift the matter of 3.5 at the end of this clause.

(Page 7, clause 3.4) — Delete '3.4'

(Page 8, clause 3.5) — Delete '3.5'.

(Page 11, clause 5.1.1, last line) — Substitute ' $\left(-\frac{n}{2}\right) q^{n-2} p^2$ ' for ' $\left(-\frac{n}{2}\right) q^{n-2} p^2$ '.

(Page 15, clause 5.6) — Substitute the following for the existing equation

$$\int_{x_1 - \frac{1}{2}}^{x_1 + \frac{1}{2}} N \left[np, \sqrt{np(1-p)} \right] dx = \int_{\frac{x_1 - np - \frac{1}{2}}{\sqrt{np(1-p)}}}^{\frac{x_1 - np + \frac{1}{2}}{\sqrt{np(1-p)}}} N(0, 1) dx$$

(Page 16, clause 6.3) — Insert the following at the end of this clause:

'Covariance — The covariance between the i^{th} and the j^{th} possible outcomes is obtained as:

$$\text{Cov}(x_i, x_j) = \sqrt{np_i p_j}$$

(Page 22, clause 8.2, line 3) — Substitute the following for the existing formula for standard deviation:

$$\sqrt{v(r)} = \left[\frac{nD}{N} \left(1 - \frac{D}{N} \right) \left(\frac{N-n}{N-1} \right) \right]^{\frac{1}{2}}$$

(Page 22, clause 8.2.1, line 5) — Substitute the following for the existing equation: .

$$\text{Standard deviation} = \sqrt{25 \left(\frac{16}{200} \right) \left(1 - \frac{16}{200} \right) \times \frac{200-25}{200-1}}$$

(Page 29, row 21, col 10) — Substitute '0 7361' for '0 7461'